

Two-Stage Fusion for Routing and Classification in Parameter-Efficient Continual Learning

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Abstract. Task-specific parameter-efficient continual learners maintain adapters whose selection affects both routing and classification. We augment HRM-PET with a random-projection (RP) head fitted by ridge regression over fixed first-task LoRA features. Its sufficient statistics are accumulated task by task without retaining examples. The same RP scores support two inference-time operations: task-level fusion with the initial adapted prediction before re-matching, and class-level fusion with the final logits using a top-two-margin gate. Across three benchmarks and four seeds, DRM-proposal accuracy increases by 1.10–2.26 points and classification accuracy by 0.69–1.50, with no consistent improvement in forgetting or backward transfer. Single-seed ablations across five backbones find little added routing benefit under ungated class fusion. Four-seed conditional contrasts show a positive routing contribution with gated class fusion on ImageNet-R and CUB-200, but an unresolved effect on CIFAR-100. The gate’s estimated contribution is negative on CUB-200, with a confidence interval including zero. A grid over the two mixing weights shows that classification accuracy is nearly flat in the routing weight, so the reported value is not a resolved optimum.

Keywords: Continual learning · Parameter-efficient tuning · Parameter routing · Random projection · Signal fusion.

1 Introduction

Continual learning asks a model to absorb a stream of tasks without losing what earlier ones taught it. Parameter-efficient tuning adapts a frozen pre-trained backbone using a small number of parameters. One important family maintains task-specific prompts or adapters and selects them at inference [5, 3]. Such methods can avoid storing old examples, but preserving parameters does not guarantee preserved accuracy.

In this family, task identity is unknown at inference and parameter selection is itself a prediction. A wrong adapter can produce unsuitable features. Larger pools, attention-weighted mixtures and explicit re-matching offer different responses to this difficulty [4, 7, 1]. We focus on HRM-PET, where routing and classification use related predictions from the adapted and unadapted backbone.

We add a complementary predictor: first-task adapter features, a frozen random projection and a ridge classifier fitted from accumulated statistics, following RanPAC [2]. Fitting uses current-task data without changing LoRA/TII training or retaining examples. Its scores revise the task proposal before re-matching and blend with final class logits, with greater weight when the primary classifier’s top-two margin narrows.

Contributions. We identify complementary errors in HRM-PET (Sect. 2.1) and introduce two inference-time fusion operations using one RP head: task-proposal fusion before re-matching and gated class fusion afterwards. The head is fitted in closed form without gradient descent. A three-dataset, four-seed study separates gains in routing/classification from unresolved retention effects. Conditional ablations and weight comparisons examine what each operation adds. We also verify a four-dataset by five-pretraining grid after the masking fix, with its remaining limitations stated in Sect. 4.

2 Related Work

Parameter pools and the selection they require. L2P [4] and Dual-Prompt [5] exemplify pool-based parameter-efficient continual learning: freeze the backbone, keep a pool of lightweight parameter sets, and pick from it at inference using a query from the bare backbone. CODA-Prompt [7] replaces the discrete pick with an attention-weighted combination; HiDe-Prompt [3] separates within-task prediction, task-identity inference and task-adaptive prediction and optimises them apart, observing that identity inference, not within-task classification, is what limits the pipeline. HRM-PET [1], our baseline, substitutes LoRA adapters [6] for prompts and adds re-matching stages that revisit the initial decision. Later work enlarges the expert family: MoTE [14] combines task-specific experts, and DLEPEM [15] combines general and adapted prototypes for matching.

Closed-form solutions in continual learning. Closed-form classifiers preserve past contributions through additive statistics of fixed features. APER [8] adapts the backbone on the first task, then uses frozen features and class prototypes. ACIL [20] uses recursive least squares, replacing a sample buffer with correlation statistics and recovering the corresponding joint solution. DS-AL [21] adds a second analytic stream. RanPAC [2] applies a fixed random projection to frozen features, accumulates second-order statistics and solves a ridge system.

Our head belongs to this family and we claim no novelty in it: the projection, the Gram accumulation and the ridge solve are RanPAC’s. What differs is its role. Rather than using the solved head as the sole predictor, we retain the adapter-routing pipeline and consult the same head at two decision points.

Correcting the classifier after the fact. Class-incremental learning often biases predictions towards recent classes [9]. iCaRL [10] uses nearest exemplar means; BiC [11] learns an affine logit correction on held-out data; Weight Aligning [12] matches new- and old-class weight norms; and LUCIR [13] normalises features and weights. Our fusion leaves the adapters, auxiliary classifier and shared head fixed. It retains no exemplars but requires current-task data to fit the RP head.

2.1 HRM-PET: The Baseline We Build On

We use the released HRM-PET [1] inference path. The seed-42 routing and classifier-overlap diagnostics below have post-fix verification; they use equal task weights, as in Sect. 4.1.

Task-identity inference (TII). The input image is passed through the *bare* backbone, with no adapter attached, to obtain a query feature $q(x) = \phi(x; \theta_{\text{ptm}})$, writing ϕ for the backbone under whatever adapter it is given. That feature enters an auxiliary classifier trained with cross-entropy, which gives unnormalised class logits $u = g(q(x); \theta_\omega)$. The initial match is then $t_0 = \mathcal{T}(\arg \max_c u_c)$.

Re-matching (DRM and CRM). The first adapted pass attaches p_{t_0} and produces $r = g_{\theta_g}(\phi(x; p_{t_0}, \theta_{\text{ptm}}))$. DRM proposes the task of $\arg \max r$ and evaluates its adapter when it differs from t_0 ; our fusion modifies this proposal, not the preceding bare TII pass. CRM selects low-confidence samples from r by the baseline’s generalised-entropy threshold and compares the current logits, under the baseline log-sum-exp score, with the adapter of the second-ranked class of r ; it is confidence-triggered, not restricted to cases where the two task identities agree. Both reuse trained modules, can require further adapter forwards, and are heuristics: neither guarantees that accuracy cannot decrease.

Shared classification head. The selected parameter set is attached to the backbone, and the shared head produces a score vector over all classes seen so far, $\ell = g(\phi(x; p_{\hat{t}}, \theta_{\text{ptm}}); \theta_g) \in \mathbb{R}^{|\mathcal{Y}|}$, predicting $\hat{y} = \arg \max_c \ell_c$, where $\hat{t}$ is the identity after DRM and CRM. Classes not yet seen are masked with $-\infty$.

Two weaknesses this paper acts on. The baseline draws its routing evidence from TII and the shared head alone. To see what a second source would be worth we built the head of Sect. 3.2 on HRM-PET’s own LoRA features and used it as a router: at seed 42 on ImageNet-R, TII routing accuracy is 63.80%, the DRM proposal 77.79% and the RP head 77.12%, while the union of TII and the head reaches 82.99%. Dividing the RP-correct/TII-wrong rate by TII’s error rate gives 53.0% recovery, against 62.0% and 46.9% on CIFAR-100 and CUB-200. These ratios use task-balanced rates, not pooled image counts. The union is an oracle,

not achieved routing accuracy, and it compares bare TII with RP rather than the actual Stage 1 inputs.

A better route need not become a better prediction: ℓ comes from the shared head, which spans every class seen so far, so a mis-routed sample can be classified correctly anyway. The RP head is moreover a complete classifier that a routing-only variant would discard: in the verified routing-only diagnostic ($w = 0.7$, $\beta = 0$) on ImageNet-R, the routed head reaches 74.31% and the projection head 70.08%, yet their union is 78.89%, the head alone contributing 4.58 task-balanced points and 2.43 and 3.87 on the other two. These measurements give the two components described next.

3 Method

3.1 Problem Statement

Class-incremental learning (CIL) learns a sequence of T tasks without forgetting earlier ones. The t -th task consists of n_t samples $\mathcal{D}_t = \{(x_i^t, y_i^t)\}_{i=1}^{n_t}$ drawn from a class set $\mathcal{C}_t$, where $x_i^t \in \mathcal{X}$ is an input image and $y_i^t \in \mathcal{Y}$ its label.

Three constraints define the setting. At step t only $\mathcal{D}_t$ is available for training. The class sets are disjoint, $\mathcal{C}_t \cap \mathcal{C}_{t'} = \emptyset$ for all $t \neq t'$, and at inference the *task identity is unknown*: the model must discriminate among all classes seen so far through a single shared classification head. Finally, this work operates in the exemplar-free setting: images from past tasks are never retained.

In the task-specific PET setting considered here [3, 1], a pre-trained backbone h_{ptm} with parameters θ_{ptm} is frozen and a lightweight parameter pool $\mathcal{P} = \{p_1, \dots, p_T\}$ is maintained, each set trained on one task; here every p_t is a LoRA adapter [6]. The shared classification head is denoted g_{θ_g} . Because the task identity is unknown at inference, the system is forced to *choose* which parameter set to attach for its primary prediction. Our fixed RP branch does not depend on this inferred task identity.

We distinguish four class-score vectors: bare TII scores u , initial adapted shared-head logits r , final re-matched logits ℓ , and RP scores s . The class-to-task map is $\mathcal{T}(\cdot)$; t_0 is the initial TII match and $\hat{t}_{\text{DRM}}$ the task proposed for direct re-matching. We write $z(\cdot)$ for per-sample standardisation with the population standard deviation floored at 10^{-6} .

3.2 The Second Source of Evidence

We fit an RP head after each task and reuse its scores in two inference-time fusion operations. HRM-PET parameters are not retrained. Figure 1 distinguishes the initial adapted logits from the bare TII scores and shows both reuse paths.

The RP head follows RanPAC [2]. Features $f \in \mathbb{R}^d$, $d = 768$, are column vectors from ViT-B/16 with the first task’s LoRA adapter (p_1 here, index 0 in code), fixed during accumulation and inference and unnormalised; changing adapters between tasks would mix incompatible spaces.

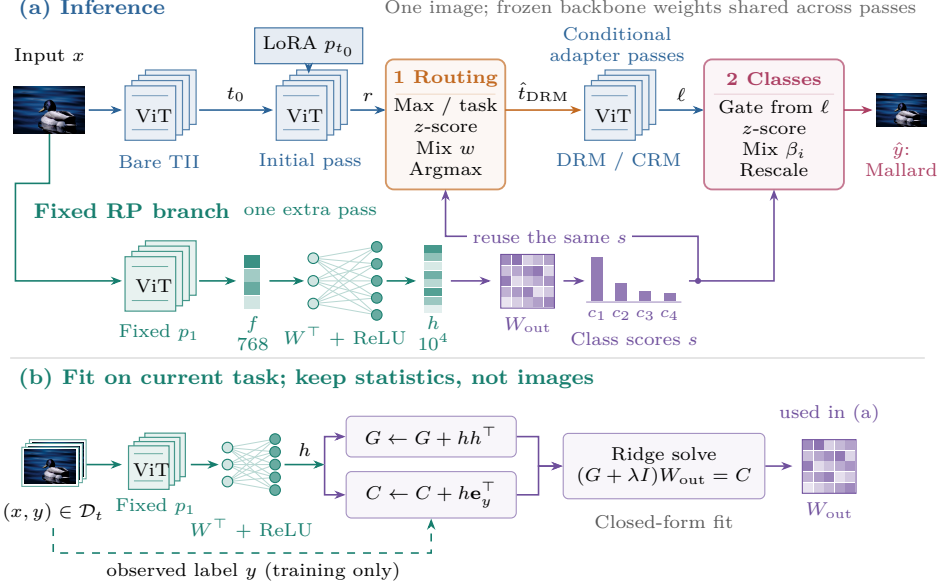


Fig. 1. Two-stage fusion. All inference passes use the same image; adapted passes share head g . TII selects t_0 ; the projection W is frozen. Stage 1 standardises task-wise maxima of r and s ; Stage 2 uses the top-two probability margin of ℓ for its continuous gate (Eqs. 6–8), then predicts by argmax. DRM runs if $\hat{t}_{\text{DRM}} \neq t_0$; CRM uses low-confidence r and its second-ranked class’s task, comparing logits by log-sum-exp (not at task 1). Only seen classes/tasks participate; score bars are schematic. Photo: T. G. Barnes/USFWS (public domain).

Projection and second-order statistics. The feature is projected into a high-dimensional space of size $M = 10^4$ and passed through a nonlinearity, and the projected features are accumulated into two matrices:

$$h = \text{ReLU}(W^T f) \in \mathbb{R}^M, \quad W \in \mathbb{R}^{d \times M}, \quad W_{jk} \sim \mathcal{N}(0, 1), \quad (1)$$

$$G = \sum_n h_n h_n^T \in \mathbb{R}^{M \times M}, \quad C = \sum_n h_n \mathbf{e}_{y_n}^T \in \mathbb{R}^{M \times |\mathcal{Y}|}, \quad (2)$$

where $\mathbf{e}_y$ is the basis vector of class y . Each current-task training sample contributes once; G and C are accumulated in float64. W is generated once with seed 1993 and then frozen.

The nonlinear expansion permits richer decision functions than a linear head on f . Without ReLU, $W^T f$ remains linear in f and adds no separating power. This does not guarantee that the expanded data become linearly separable.

Classification by ridge regression. The output weights are the closed-form solution of a regularised least-squares problem:

$$W_{\text{out}} = (G + \lambda I)^{-1} C \in \mathbb{R}^{M \times |\mathcal{Y}|}, \quad s = W_{\text{out}}^T h \in \mathbb{R}^{|\mathcal{Y}|}. \quad (3)$$

The implementation solves the regularised system rather than inverting it.

Two inherited properties. Both G and C in (2) are *sums* over samples, so for a *fixed* feature map they, and with them W_{out} , do not depend on the order in which samples arrived: nothing an earlier task contributed is overwritten, and the head after T tasks equals the ridge solution of a joint fit over every feature seen. Their sizes do not grow with the sample count, and no per-sample feature is retained. This exact-arithmetic identity does not imply bitwise order invariance in floating point or an absence of forgetting: new classes can outscore old ones despite unchanged statistics. Moreover, changing the first task changes its adapter and hence the feature map.

3.3 Two-Stage Fusion

Both stages blend sources with different score scales. Direct mixing can make the nominal weights poorly reflect their influence. Each source is standardised per sample:

$$z(v) = \frac{v - \text{mean}(v)}{\sigma_\epsilon(v)}, \quad \sigma_\epsilon(v) = \max\{\text{std}(v), 10^{-6}\}, \quad (4)$$

where std is the population standard deviation. Moments are over seen tasks in Stage 1 and seen classes in Stage 2, excluding unseen entries.

First stage: fusion at the routing level. Stage 1 runs after the initial adapted pass and before DRM. Its input is r , the shared-head logits under the adapter selected by bare TII, not auxiliary logits u . Both r and RP scores s are mapped to seen tasks by $r_t^{\text{task}} = \max_{c \in \mathcal{C}_t} r_c$ and $s_t^{\text{task}} = \max_{c \in \mathcal{C}_t} s_c$ for $t = 1, \dots, \hat{T}$, with $\hat{T}$ the number of tasks seen so far; classes of unseen tasks never enter either maximum. The two vectors are blended after standardisation:

$$\text{fused} = w z(r^{\text{task}}) + (1 - w) z(s^{\text{task}}), \quad \hat{t}_{\text{DRM}} = \arg \max_t \text{fused}_t, \quad w = 0.7. \quad (5)$$

For one seen task the standard deviation is zero, so the floor keeps the operation finite. The fused task replaces the baseline DRM proposal $\mathcal{T}(\arg \max r)$; the initial TII match, CRM candidate rule, confidence test and shared head are unchanged. At $w = 1$ the ordering is that of r^{task} , recovering that proposal; the full baseline also needs $\beta = 0$.

Second stage: gated fusion at the classification level. Because ℓ comes from the shared head, it can predict the correct class in spite of a wrong routing decision, so we use the projection head itself as a secondary classifier, this time taking its class scores s . A fixed blend would alter confident primary predictions;

we instead reduce the RP share as the top-two probability margin grows, while recognising that confident predictions can still be wrong:

$$\beta_i = \beta \left(1 - \frac{p_{(1)} - p_{(2)}}{p_{(1)} + p_{(2)}} \right), \quad \beta = 0.5, \quad (6)$$

where $p_{(1)} \geq p_{(2)}$ are the two largest probabilities of $\text{softmax}(\ell)$ over the classes seen so far. Dividing by the sum makes the quantity depend only on the ratio $p_{(2)}/p_{(1)}$, so the gate measures how threatened the top class is rather than how much mass the pair holds. The RP share falls to zero as the top class dominates and reaches β when the two are equal: a continuous weight, not a binary test.

The blend reuses (4) and then maps the result back onto the scale of ℓ :

$$m_i = (1 - \beta_i) z(\ell_i) + \beta_i z(s_i), \quad (7)$$

$$\hat{\ell}_i = m_i \cdot \sigma_\epsilon(\ell_i) + \text{mean}(\ell_i) \in \mathbb{R}^{|\mathcal{Y}|}. \quad (8)$$

The positive affine map-back preserves rankings and restores a reference scale for cross-entropy, not a calibration guarantee: for standardised nondegenerate vectors with correlation ρ and the floor inactive, the mixture variance is $(1 - \beta_i)^2 + \beta_i^2 + 2\beta_i(1 - \beta_i)\rho$, which need not equal one. At $\beta = 0$ this stage leaves ℓ unchanged.

Cost. At inference the head costs one extra adapter forward per sample, serving both stages. That is the head’s own cost, not the pipeline’s difference, which we did not measure: changing the DRM proposal can change the number of adapter evaluations.

Per task the head accumulates G and C over the current task’s training split and solves one ridge system, so its footprint is set by M , not by how many samples have been seen. At $M = 10^4$ the Gram matrix holds 10^8 float64 entries, or 800 MB. Together with its regularised copy and Cholesky factor, these three matrices require approximately 2.4 GB; this is an analytical estimate, not a measured total-memory peak. A historical ten-task evaluation on ImageNet-R took 283 s on one RTX 4090 at batch 24; neither its phase-level costs nor its overhead over the baseline were measured.

4 Experiments

4.1 Setup

The backbone is ViT-B/16 [22], pre-trained on ImageNet-21K. Rank-8 LoRA modifies the key/value projections in the first five blocks. The four-seed study splits ImageNet-R [23], CIFAR-100 [24] and CUB-200-2011 [25] into 10 tasks of 20, 10 and 20 classes, respectively. The grid replaces CUB with ImageNet-A [26] (10 $\times$ 20 classes) and the five-dataset protocol of [1] (one domain per task). Its pre-training settings are Sup-21K, MoCo-v3 [27], iBOT-1K/iBOT-21K [28] and DINO [29].

Each seed (42–45) has separately trained LoRA/TII parameters; pre-trained backbone weights remain frozen. Both methods use identical saved checkpoints and evaluation paths. RP fits current-task training images once using the evaluation transform: resize to 256, centre crop to 224, and convert to a tensor; CIFAR additionally uses its configured channel normalisation. No test labels enter the fit. We use $M = 10^4$, ReLU, $\lambda = 10^4$, no additional RP feature normalisation or temperature calibration, and no fusion ramp. Fusion starts at task 1; CRM retains **En=gen**, threshold -10 , and the second-ranked class candidate. Training/evaluation scripts and dataset preprocessing are versioned in the code repository [30].

Acc@task in the released evaluator measures the DRM task proposal. In the default class-based routing path, CRM can change the output logits without updating the task identity passed to this metric; it is therefore not a measure of final adapter selection after CRM. Acc@1, Acc@5 and Loss use the final classification logits. Final metrics average the per-task values with equal task weights, not by pooling all test images. Writing $a_{i,t}$ for classification accuracy on task i at stage t , the retention measures are

$$\text{Forgetting} = \frac{1}{T-1} \sum_{i=1}^{T-1} \left(\max_{i \leq t \leq T} a_{i,t} - a_{i,T} \right), \quad (9)$$

$$\text{Backward} = \frac{1}{T-1} \sum_{i=1}^{T-1} (a_{i,T} - a_{i,i}). \quad (10)$$

Both averages exclude the last task, and the maximum includes only stages at which task i has been learned. Lower is better for Loss and Forgetting.

Sanity check. $w = 1$ restores the baseline DRM proposal and $\beta = 0$ bypasses class fusion. After correcting unseen-class masking, this identity matches the baseline to logged precision at every stage for all three datasets and seeds 42–45. For example, ImageNet-R seed 42 gives Acc@task 77.7914, Acc@1 74.0191 and Loss 1.2305. Main runs, four-seed controls and the joint weight grid were re-evaluated. Backbone/first-adapter feature tensors match across checkpoints in all main/grid checkpoint sets; all forty grid evaluations are verified.

4.2 Main Results

Table 1 reports all three datasets over four seeds. The change column is pairwise — taken per seed, then averaged — so its standard deviation describes the stability of the improvement, not the seeds’ difficulty.

Intervals are paired 95% Student- t , $\bar{\Delta} \pm 3.182 s_{\Delta}/\sqrt{4}$: imprecise at four seeds, and assuming approximately normal seed-wise differences. Correlated metrics mean we discuss axes rather than count confirmations.

Table 1. Final results, mean $\pm$ sample SD over seeds 42–45; Change is paired full minus baseline. Accuracy/retention are percentages (changes in points). All three datasets have complete post-fix verification.

Dataset	Metric	HRM-PET	Proposed	Change
<i>ImageNet-R</i>	Acc@task	77.58 ± 0.33	79.85 ± 0.18	$+2.264 \pm 0.158$
	Acc@1	73.94 ± 0.48	75.18 ± 0.24	$+1.236 \pm 0.416$
	Acc@5	86.71 ± 0.21	88.02 ± 0.17	$+1.312 \pm 0.148$
	Loss	1.23 ± 0.01	1.19 ± 0.01	-0.040 ± 0.002
	Forgetting	3.36 ± 0.56	3.30 ± 0.35	-0.056 ± 0.334
	Backward	-3.16 ± 0.61	-3.20 ± 0.42	-0.047 ± 0.282
<i>CIFAR-100</i>	Acc@task	89.77 ± 0.11	90.87 ± 0.09	$+1.103 \pm 0.066$
	Acc@1	89.78 ± 0.03	90.47 ± 0.05	$+0.690 \pm 0.063$
	Acc@5	98.65 ± 0.03	98.86 ± 0.07	$+0.218 \pm 0.057$
	Loss	0.39 ± 0.00	0.39 ± 0.00	-0.0003 ± 0.0012
	Forgetting	3.89 ± 0.08	3.74 ± 0.12	-0.150 ± 0.095
	Backward	-3.87 ± 0.07	-3.74 ± 0.11	$+0.136 \pm 0.095$
<i>CUB-200</i>	Acc@task	93.12 ± 0.14	94.34 ± 0.10	$+1.226 \pm 0.149$
	Acc@1	86.38 ± 0.25	87.88 ± 0.27	$+1.495 \pm 0.353$
	Acc@5	97.04 ± 0.10	97.44 ± 0.13	$+0.400 \pm 0.103$
	Loss	0.57 ± 0.00	0.53 ± 0.01	-0.036 ± 0.004
	Forgetting	2.54 ± 0.35	2.18 ± 0.12	-0.359 ± 0.325
	Backward	-2.42 ± 0.39	-2.14 ± 0.13	$+0.279 \pm 0.391$

Routing (Acc@task): improved. The gain is +1.10 to +2.26 points, with positive paired intervals on all three datasets. This is the axis the first stage acts on directly.

Classification (Acc@1, Acc@5): improved. Acc@1 rises by +0.69 to +1.50 points and Acc@5 by +0.22 to +1.31, with positive paired intervals for both metrics on all three datasets. The full method’s Acc@1 improvement repeats across every seed. These aggregate gains do not establish how many routing repairs also repair classification.

Retention (Forgetting, Backward): unresolved. All six paired retention intervals include zero. Corrected CIFAR Forgetting is $-0.150 [-0.302, +0.002]$ and Backward $+0.136 [-0.015, +0.287]$. These related metrics are not independent confirmations. Neither consistent improvement nor equivalence is established: four seeds leave substantial uncertainty, and no equivalence margin was specified. The single-seed decomposition in Sect. 4.4 is descriptive.

4.3 Placement Against Prior Methods

Table 2 places the method among the baselines the original work reports. Its first eight rows are quoted from Table 1 of [1] under Sup-21K, averages over

Table 2. Final average accuracy (%) under Sup-21K. Rows above the rule are quoted from Table 1 of [1], averaged over three seeds; the two below are ours at seed 42, from a different implementation and seed set. Bold indicates the best value within each group, not a controlled cross-group ranking. S-Prompt++ and HiDe-LoRA are the variants evaluated in [1].

Method	CIFAR-100	ImageNet-R	ImageNet-A	5-Datasets
DualPrompt [5]	86.60	68.79	39.76	76.09
S-Prompt++ [16]	88.81	69.68	39.10	83.38
CODA-Prompt [7]	86.94	70.03	30.96	62.54
LAE [17]	85.10	72.25	40.36	64.16
CPrompt [18]	83.18	69.24	37.60	71.72
InfLoRA [19]	85.28	72.24	42.12	71.97
HiDe-LoRA [3]	88.43	71.90	42.18	93.25
HRM-PET [1]	89.45	73.86	44.28	93.99
HRM-PET, our run	89.77	74.02	45.85	91.37
Proposed	90.53	75.51	47.42	93.77

three seeds; the last two are our own runs at seed 42 through the code path of Sect. 4.1.

Our seed-42 baseline differs from the published mean by +0.32, +0.16, +1.57 and −2.62 points on the four datasets, respectively. These reproduction gaps preclude a controlled comparison across the two groups. The controlled comparison remains the paired baseline–full difference in Table 1.

4.4 The Full Grid

The 4×5 grid uses one code path at seed 42. Post-fix verification of all forty evaluations reproduces final accuracy and Loss; ten full-method runs have changed retention values. All twenty cells improve Acc@task by +0.97 to +4.57 and Acc@1 by +0.59 to +3.58, against baseline Acc@1 of 31.8–93.8. Verified final accuracy on earlier tasks rises in every cell (+1.83 on average), versus +1.58 at learning time. Backward, their difference, worsens in five cells: learning-time accuracy rose more, not final accuracy less. This decomposition uses rounded per-task logs and remains descriptive.

4.5 Ablation

The five-backbone ImageNet-R ablation uses seed 42; baseline Acc@1 ranges from 68.47 to 74.02. We report differences between specified variants.

Routing with ungated class fusion. Verified routing-only and ungated class-only variants improve Acc@1 on each backbone by +0.29–+0.73 and +0.51–+3.07, respectively. Adding routing to ungated class fusion changes Acc@1 by −0.07 to +0.02, not establishing equivalence. All fifteen ablation results have post-fix verification; some retention values changed, but final accuracy did not.

Post-fix full-minus-gated-class-only Acc@1 is +0.13, +0.29, +0.78, +0.43 and +0.77 on Sup-21K, MoCo-v3, iBOT-1K, iBOT-21K and DINO at seed 42. These single-seed gains establish neither reliability nor retention benefit: MoCo-v3 Forgetting rises from 3.44 to 3.69. The gate is not uniformly beneficial either: full minus ungated full is -0.26 Acc@1 on MoCo-v3 and -0.54 on iBOT-1K.

Conditional contributions over four seeds. For all three datasets and seeds 42–45, class-only fusion uses $w = 1$, $\beta = 0.5$, with or without the margin gate. Full minus gated class-only measures conditional routing; gated minus ungated class-only measures the gate without routing fusion. These are not independent stage effects.

The conditional routing effect is +0.083 [+0.026, +0.140] on ImageNet-R, -0.030 $[-0.085, +0.025]$ on CIFAR-100 and +0.156 [+0.055, +0.257] on CUB-200. The gate is +0.603 and +0.355 on the first two, both excluding zero, and -0.284 $[-0.570, +0.002]$ on CUB-200. Each pair sums to the full-against-ungated-class-only change, +0.69, +0.33 and -0.13 , to within 0.005. Routing is positive on two datasets and unresolved on CIFAR. The gate’s CUB estimate is negative but its interval includes zero; this does not establish harm. Full still improves over HRM-PET in Table 1. Only the first fusion stage changes Acc@task.

4.6 Sensitivity to the Mixing Weights

The two weights were fixed sequentially, w before the second stage existed and β afterwards, so the second stage may have shifted the optimum of the first. Table 3 reports Acc@1 over the joint grid on ImageNet-R at seed 42 with the margin gate active.

Table 3. Acc@1 over the $w \times \beta$ grid, ImageNet-R at seed 42, full method with the margin gate.

	$\beta = 0.3$	$\beta = 0.5$	$\beta = 0.7$
$w = 0.6$	75.28	75.61	75.29
$w = 0.7$ (reported)	75.22	75.51	75.24
$w = 0.8$	75.12	75.50	75.27

The grid favours $\beta = 0.5$ at every w ; its maximum is $w = 0.6$, not the reported 0.7. At $\beta = 0.5$, Acc@1 spans 0.11 points, versus about one point for Acc@task. Across four seeds, 0.6 gains +0.332 [+0.143, +0.522] Acc@task and +0.108 $[-0.064, +0.280]$ Acc@1 over 0.7 with the gate and $\beta = 0.5$; the Acc@1 sign reverses on seed 43. This contrast and all nine cells were verified after the masking fix: final Acc@1 is unchanged, but some intermediate/retention values differ. These full-method comparisons establish neither equivalence nor routing-only optimality. We retain the earlier $w = 0.7$ for consistency, not as a known optimum.

4.7 Limitations

ImageNet-R and CUB informed development; neither is an untouched holdout. More seeds do not remove selection bias. CUB/pre-training overlap [25] was not audited. The grids and five-backbone ablation are single-seed; only the three-dataset contrasts use four seeds. CIFAR’s conditional routing effect is unresolved, and gate benefits vary across backbones. Same-feature RP-only reaches mean Acc@1 69.89/88.80/87.26 on ImageNet-R/CIFAR/CUB, below the full method, but is not a same-protocol RanPAC reproduction. Quoted baselines use another implementation; our 5-Datasets result is below the published figure. Retention benefit/equivalence remains unestablished; phase-level cost and actual full-fusion repair/harm counts were not measured. The loader skips two zero-byte NotMNIST training files, also empty in the local source archive; we did not replace them.

5 Conclusion

A fixed-feature RP head supports task-proposal and class fusion, gaining 0.69–1.50 Acc@1 points across three datasets. Gated routing effects are dataset-dependent; the negative CUB gate estimate remains unresolved. Main results, oracle diagnostics, ablations and the cross-dataset grid have post-fix verification. Accuracy gains do not establish reduced forgetting.

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References

1. Wang, W., Jia, G., Liu, X., Lin, L., Yang, J.: Hybrid re-matching for continual learning with parameter-efficient tuning. In: NeurIPS (2025)
2. McDonnell, M.D., Gong, D., Parvaneh, A., Abbasnejad, E., van den Hengel, A.: RanPAC: random projections and pre-trained models for continual learning. In: NeurIPS (2023)
3. Wang, L., Xie, J., Zhang, X., Huang, M., Su, H., Zhu, J.: Hierarchical decomposition of prompt-based continual learning: rethinking obscured sub-optimality. In: NeurIPS (2023)
4. Wang, Z., Zhang, Z., Lee, C.Y., Zhang, H., Sun, R., Ren, X., Su, G., Perot, V., Dy, J., Pfister, T.: Learning to prompt for continual learning. In: CVPR (2022)
5. Wang, Z., Zhang, Z., Ebrahimi, S., Sun, R., Zhang, H., Lee, C.Y., Ren, X., Su, G., Perot, V., Dy, J., et al.: DualPrompt: complementary prompting for rehearsal-free continual learning. In: ECCV (2022)
6. Hu, E.J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., Chen, W.: LoRA: low-rank adaptation of large language models. In: ICLR (2022)
7. Smith, J.S., Karlinsky, L., Gutta, V., Cascante-Bonilla, P., Kim, D., Arbelle, A., Panda, R., Feris, R., Kira, Z.: CODA-Prompt: continual decomposed attention-based prompting for rehearsal-free continual learning. In: CVPR (2023)
8. Zhou, D.W., Cai, Z.W., Ye, H.J., Zhan, D.C., Liu, Z.: Revisiting class-incremental learning with pre-trained models: generalizability and adaptivity are all you need. *Int. J. Comput. Vis.* **133**, 1012–1032 (2025)
9. Masana, M., Liu, X., Twardowski, B., Menta, M., Bagdanov, A.D., van de Weijer, J.: Class-incremental learning: survey and performance evaluation on image classification. *IEEE Trans. Pattern Anal. Mach. Intell.* **45**(5), 5513–5533 (2023)
10. Rebuffi, S.A., Kolesnikov, A., Sperl, G., Lampert, C.H.: iCaRL: incremental classifier and representation learning. In: CVPR (2017)
11. Wu, Y., Chen, Y., Wang, L., Ye, Y., Liu, Z., Guo, Y., Fu, Y.: Large scale incremental learning. In: CVPR (2019)
12. Zhao, B., Xiao, X., Gan, G., Zhang, B., Xia, S.T.: Maintaining discrimination and fairness in class incremental learning. In: CVPR (2020)
13. Hou, S., Pan, X., Loy, C.C., Wang, Z., Lin, D.: Learning a unified classifier incrementally via rebalancing. In: CVPR (2019)
14. Li, L., Wu, Z., Ji, Y.: MoTE: mixture of task-specific experts for pre-trained model-based class-incremental learning. *Knowledge-Based Systems* **324**, 113795 (2025). <https://doi.org/10.1016/j.knosys.2025.113795>
15. Zhao, H., Liu, R., Liu, Y.: Dynamic LoRA-experts and prototype-ensemble matching for class-incremental learning. *Applied Sciences* **16**(12), 6153 (2026)
16. Wang, Y., Huang, Z., Hong, X.: S-Prompts learning with pre-trained transformers: an Occam’s razor for domain incremental learning. In: NeurIPS (2022)
17. Gao, Q., Zhao, C., Sun, Y., Xi, T., Zhang, G., Ghanem, B., Zhang, J.: A unified continual learning framework with general parameter-efficient tuning. In: ICCV (2023)
18. Gao, Z., Cen, J., Chang, X.: Consistent prompting for rehearsal-free continual learning. In: CVPR (2024)
19. Liang, Y.S., Li, W.J.: InfLoRA: interference-free low-rank adaptation for continual learning. In: CVPR (2024)
20. Zhuang, H., Weng, Z., Wei, H., Xie, R., Toh, K.A., Lin, Z.: ACIL: analytic class-incremental learning with absolute memorization and privacy protection. In: NeurIPS (2022)

21. Zhuang, H., He, R., Tong, K., Zeng, Z., Chen, C., Lin, Z.: DS-AL: a dual-stream analytic learning for exemplar-free class-incremental learning. In: AAAI, vol. 38(15), pp. 17237–17244 (2024). <https://doi.org/10.1609/aaai.v38i15.29670>
22. Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., et al.: An image is worth 16x16 words: transformers for image recognition at scale. In: ICLR (2021). <https://arxiv.org/abs/2010.11929>
23. Hendrycks, D., Basart, S., Mu, N., Kadavath, S., et al.: The many faces of robustness: a critical analysis of out-of-distribution generalization. In: ICCV (2021). <https://github.com/hendrycks/imagenet-r>
24. Krizhevsky, A.: Learning multiple layers of features from tiny images. Technical report, University of Toronto (2009). <https://www.cs.toronto.edu/~kriz/cifar.html>
25. Wah, C., Branson, S., Welinder, P., Perona, P., Belongie, S.: The Caltech-UCSD Birds-200-2011 dataset. Technical report CNS-TR-2011-001, California Institute of Technology (2011). https://www.vision.caltech.edu/datasets/cub_200_2011/
26. Hendrycks, D., Zhao, K., Basart, S., Steinhardt, J., Song, D.: Natural adversarial examples. In: CVPR, pp. 15262–15271 (2021)
27. Chen, X., Xie, S., He, K.: An empirical study of training self-supervised vision transformers. In: ICCV, pp. 9640–9649 (2021)
28. Zhou, J., Wei, C., Wang, H., Shen, W., Xie, C., Yuille, A., Kong, T.: iBOT: image BERT pre-training with online tokenizer. In: ICLR (2022). <https://arxiv.org/abs/2111.07832>
29. Caron, M., Touvron, H., Misra, I., Jégou, H., Mairal, J., Bojanowski, P., Joulin, A.: Emerging properties in self-supervised vision transformers. In: ICCV, pp. 9650–9660 (2021)
30. Trường, N.T., Thái, T.Q.: Two-stage fusion for PET-based continual learning: implementation, paper-system revision 4989ab8 (2026). <https://github.com/Truong140206/Two-Stage-Fusion-for-PET-Based-Continual-Learning->